PETSc vs MATLAB 3D SSR Performance Comparison

slope_stability

Indirect SSR, P2 elements, MPI-8 PETSc vs OMP-8 MATLAB

Repository:	`/home/beremi/repos/slope_stability-1`
Study root:	`docs/petsc_matlab_performance_comparison`
Raw artifacts:	`artifacts/petsc_matlab_performance_comparison`

Contents

1 Methodology

The comparison uses indirect SSR only and P2 elements only. The three primary cases are homogeneous 3D SSR, heterogeneous 3D SSR, and heterogeneous 3D seepage SSR on the concave mesh ladder.

PETSc runs use `mpirun-n8` with `OMP_NUM_THREADS=1`. MATLAB runs use `OMP_NUM_THREADS=8` and the BoomerAMG MATLAB path with 8 threads. All main PETSc and MATLAB runs for a given case share the same continuation horizon, estimated by a PETSc smoke run on the coarsest physical mesh with `d_lambda_diff_scaled_min=0.01`. Main runs then disable the continuation delta-lambda stop and terminate on $\omega_{\max}$. The seepage case also uses a case-specific continuation seed `lambda_init=0.00546875` and `d_lambda_init=0.01` in both codes to avoid repeated failed initialization solves on the waterlevels ladder.

The heterogeneous 3D case adds one appendix variant where PETSc Newton stopping changes from relative residual to absolute delta-lambda with tolerance 10^{-4}.

The homogeneous and heterogeneous dry cases use the requested relative-residual Newton tolerance 10^{-4}. For the waterlevels seepage ladder, repeated smoke attempts showed that the original 10^{-4} init-phase target was not attainable on the chosen meshes with the shared study formulation, so the seepage comparison uses the same relative-residual criterion in both codes with a case-specific tolerance 5×10^{-2}. This keeps PETSc and MATLAB aligned within the seepage case while preserving the original 10^{-4} setting for the two dry reference cases.

2 Environment And Outputs

The report consumes only normalized CSV files under `data/`. Figures are generated as vector PDFs under `figures/`, and table fragments are generated under `generated/`. Long-running raw solver outputs remain under `artifacts/petsc_matlab_performance_comparison`. Figures and tables include only completed PETSc/MATLAB runs captured in those CSV files.

3 Homogeneous 3D SSR

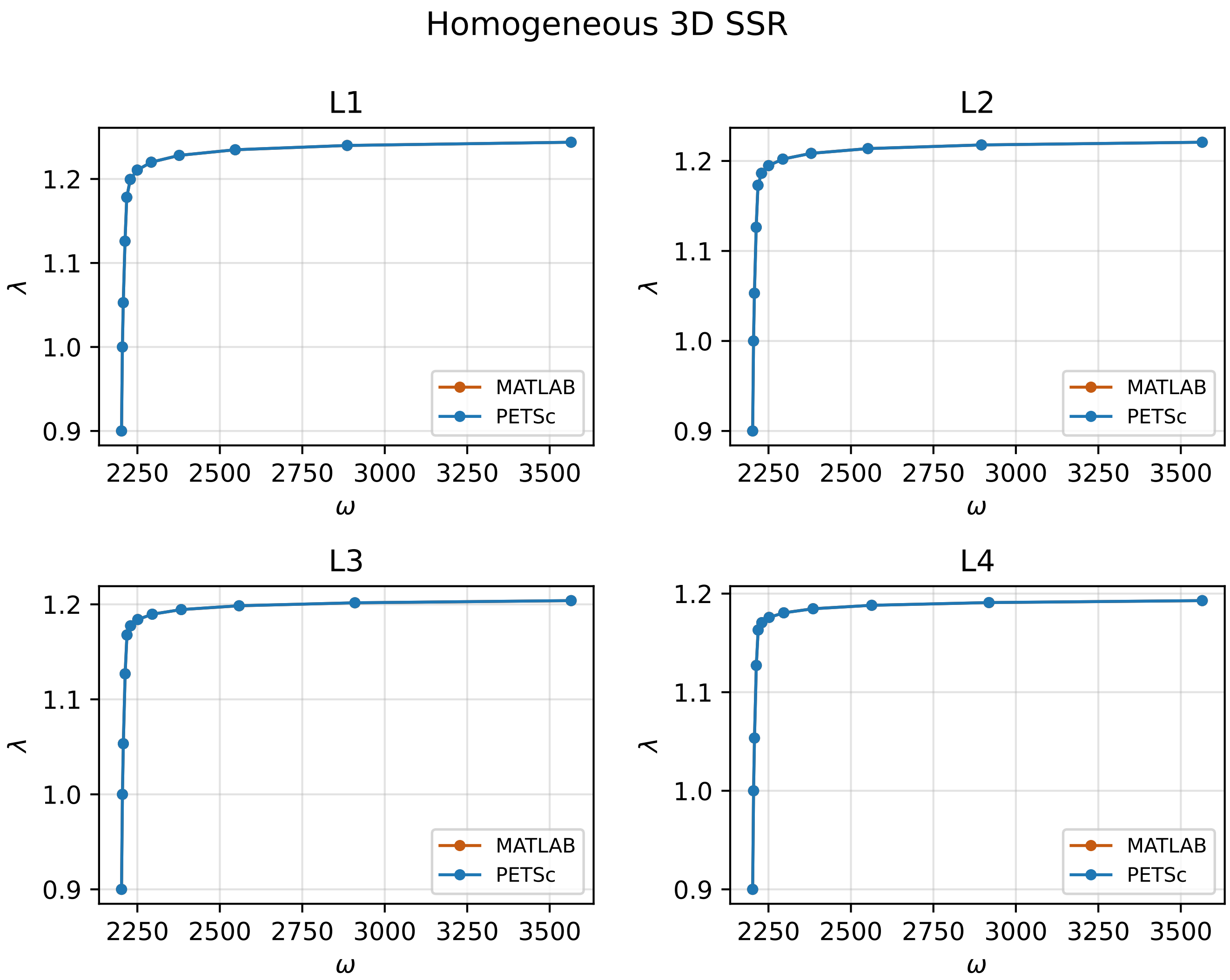

Figure 1: Continuation curves by mesh level for the homogeneous 3D SSR case.

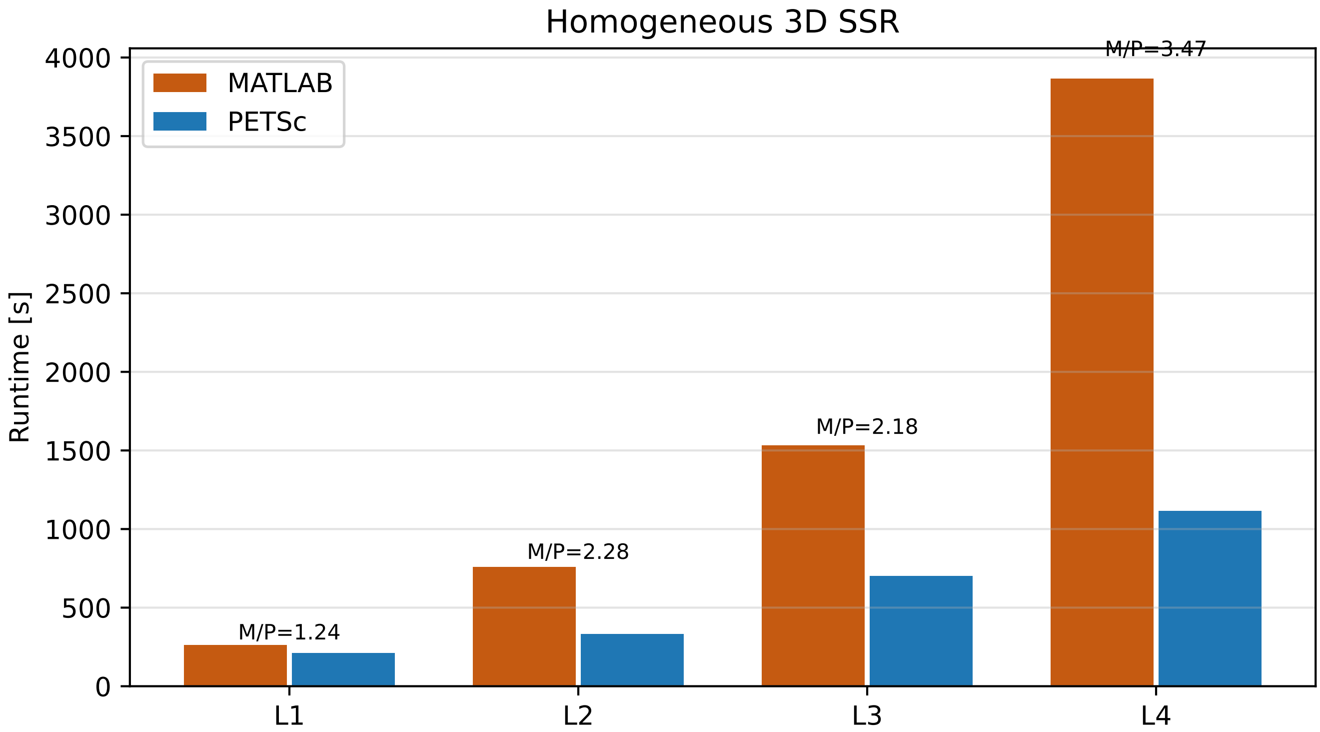

Figure 2: Runtime comparison by mesh level for the homogeneous 3D SSR case.

Table 1: Homogeneous 3D SSR summary.

Level	Unknowns	PETSc [s]	MATLAB [s]	M/P	PETSc steps	MATLAB steps	Stop (P/M)
L1	71947	210.487	261.782	1.24	12	12	omega max / omega max
L2	138382	332.843	759.187	2.28	12	12	omega max / omega max
L3	264607	701.102	1531.512	2.18	12	12	omega max / omega max
L4	508962	1115.231	3865.076	3.47	12	12	omega max / omega max

4 Heterogeneous 3D SSR

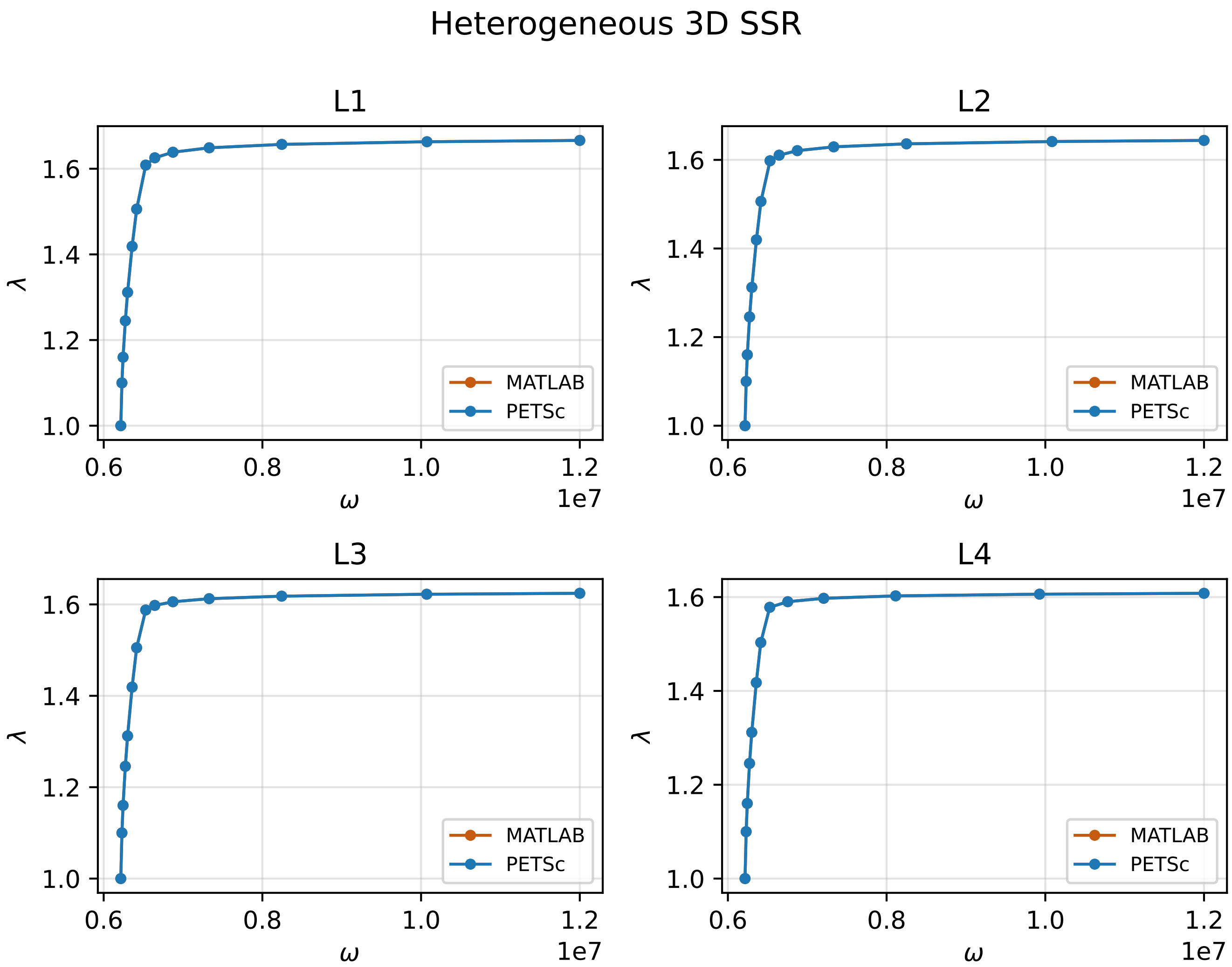

Figure 3: Continuation curves by mesh level for the heterogeneous 3D SSR case.

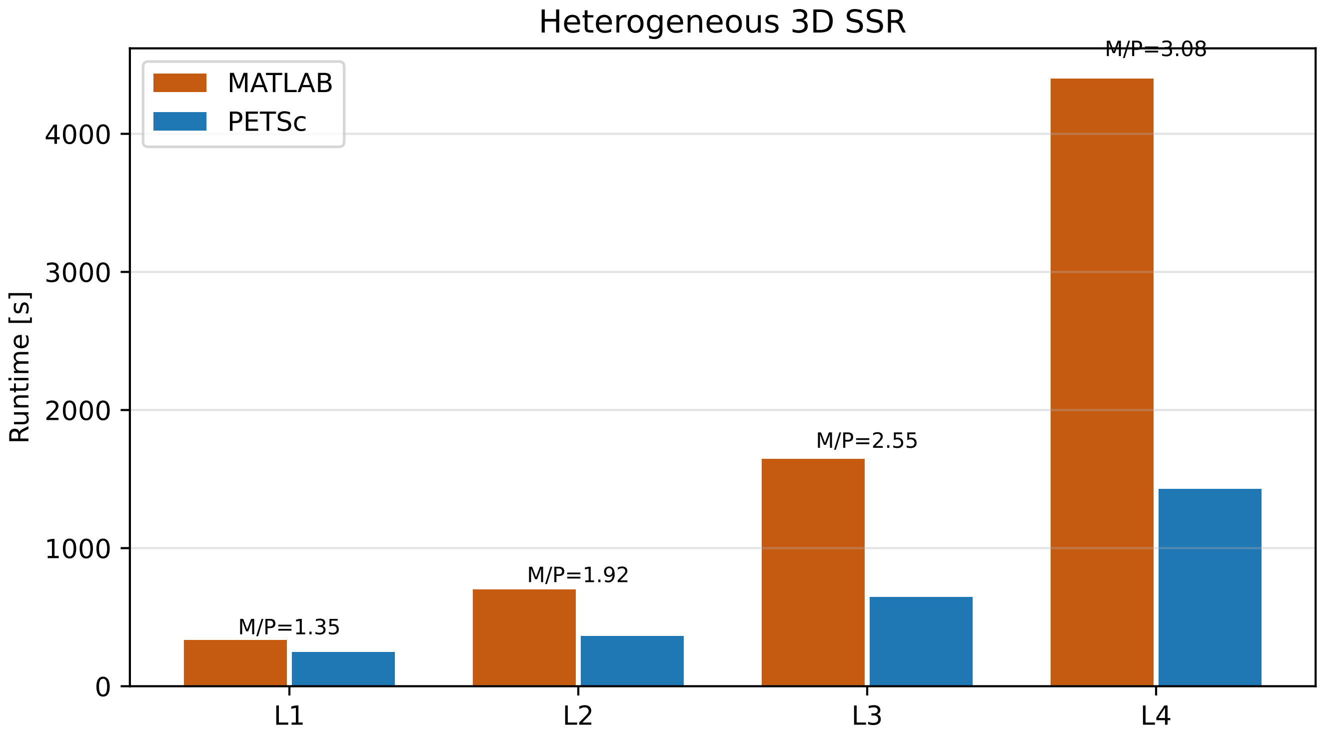

Figure 4: Runtime comparison by mesh level for the heterogeneous 3D SSR case.

Table 2: Heterogeneous 3D SSR summary.

Level	Unknowns	PETSc [s]	MATLAB [s]	M/P	PETSc steps	MATLAB steps	Stop (P/M)
L1	80362	247.304	334.248	1.35	14	14	omega max / omega max
L2	145298	365.262	700.772	1.92	14	14	omega max / omega max
L3	265739	645.949	1645.273	2.55	14	14	omega max / omega max
L4	490015	1430.224	4399.089	3.08	13	13	omega max / omega max

5 Heterogeneous 3D Seepage SSR

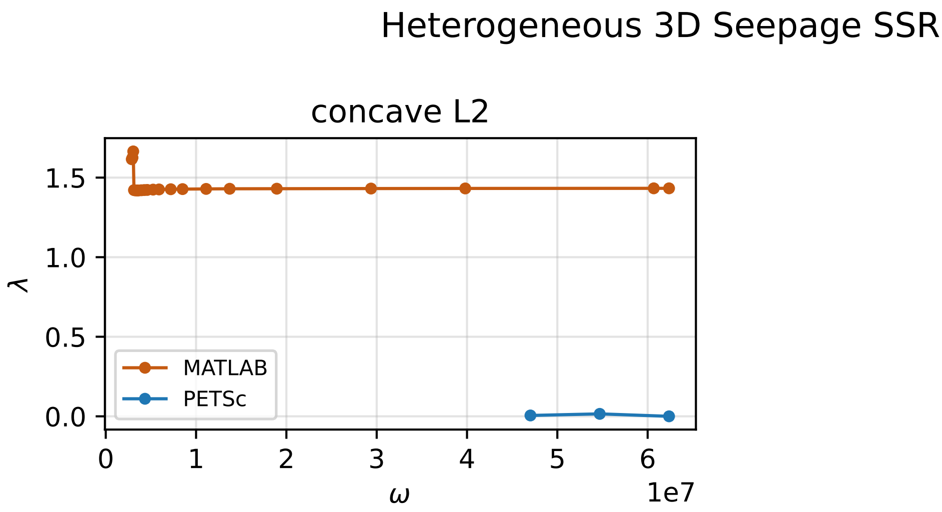

Figure 5: Continuation curves by mesh level for the heterogeneous 3D seepage SSR case.

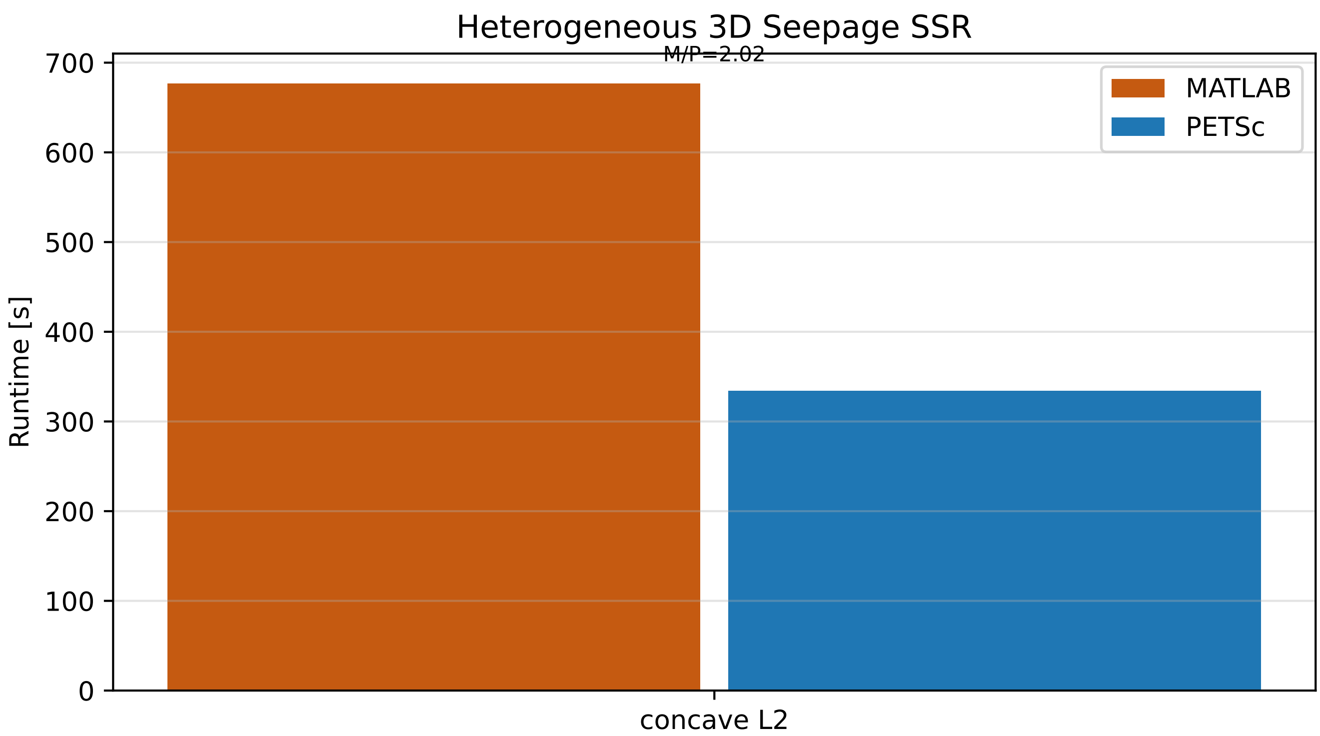

Figure 6: Runtime comparison by mesh level for the heterogeneous 3D seepage SSR case.

Table 3: Heterogeneous 3D Seepage SSR summary.

Level	Unknowns	PETSc [s]	MATLAB [s]	M/P	PETSc steps	MATLAB steps	Stop (P/M)
concave L2	205080	334.116	676.333	2.02	3	21	omega max / omega max

Only the `concave_L2` seepage level completed under the requested mixed PMG-shell study setup. The next seepage level, `concave`, reproducibly failed on the PETSc side after hierarchy construction and outer-solver setup, with a rank-4 SIGSEGV in the mixed `P2(LN)->P1(LN)->P1(Lc)` shell hierarchy. Same-mesh fallback diagnostics avoided the crash but did not yield a usable converged benchmark within the locked study protocol, so the seepage section reports the completed `concave_L2` comparison only.

A Heterogeneous 3D Newton Stopping Appendix

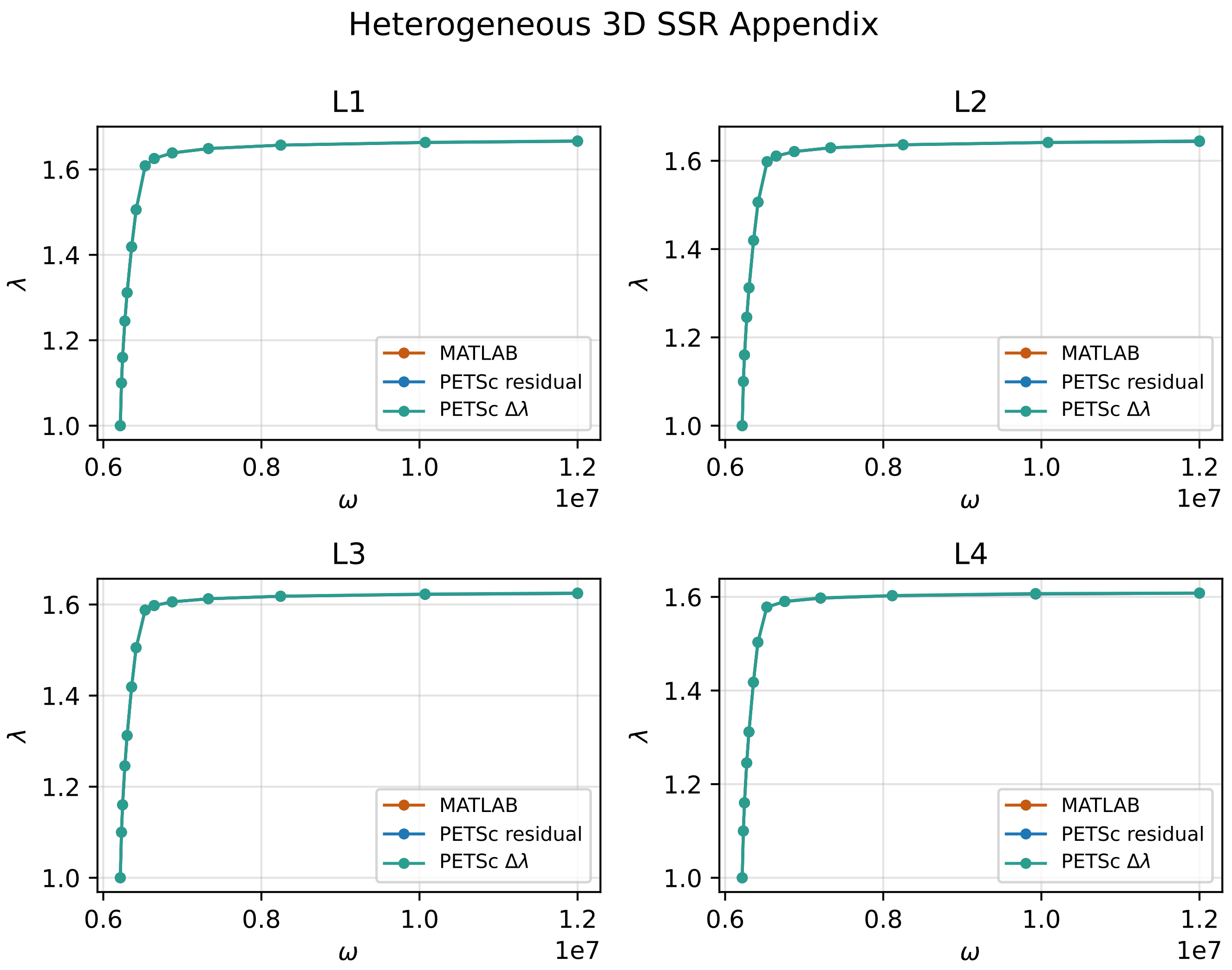

Figure 7: Continuation curves for heterogeneous 3D SSR with the PETSc delta-lambda appendix variant.

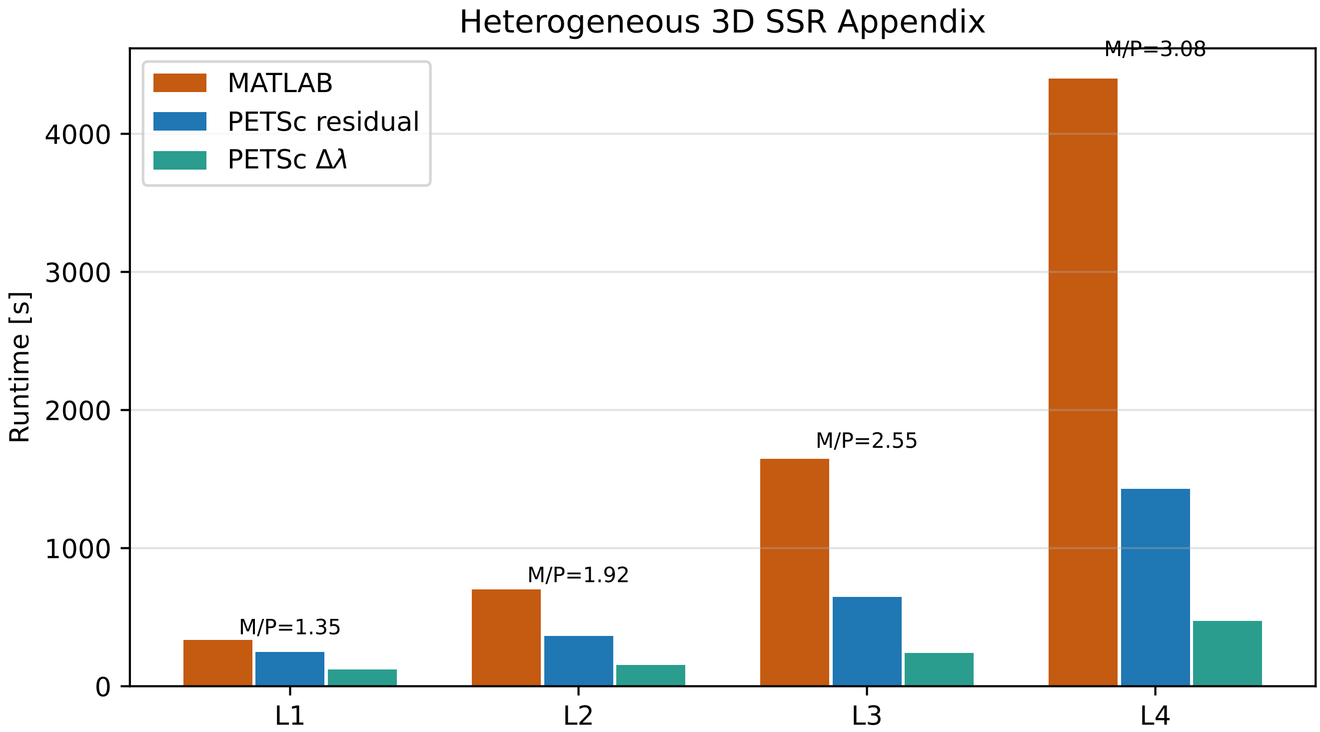

Figure 8: Timing comparison for heterogeneous 3D SSR with the PETSc delta-lambda appendix variant.

Table 4: Heterogeneous 3D SSR appendix: PETSc residual-stop vs PETSc delta-lambda-stop.

Level	PETSc res. [s]	PETSc $\Delta\lambda$ [s]	MATLAB [s]	Res. steps	$\Delta\lambda$ steps	MATLAB steps
L1	247.304	122.460	334.248	14	14	14
L2	365.262	152.865	700.772	14	14	14
L3	645.949	239.822	1645.273	14	14	14
L4	1430.224	473.999	4399.089	13	13	13